AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

1.Create an EC2 Windows Instance in AWS

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Objective

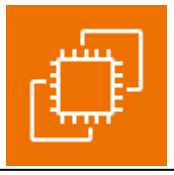
EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Windows Instances:

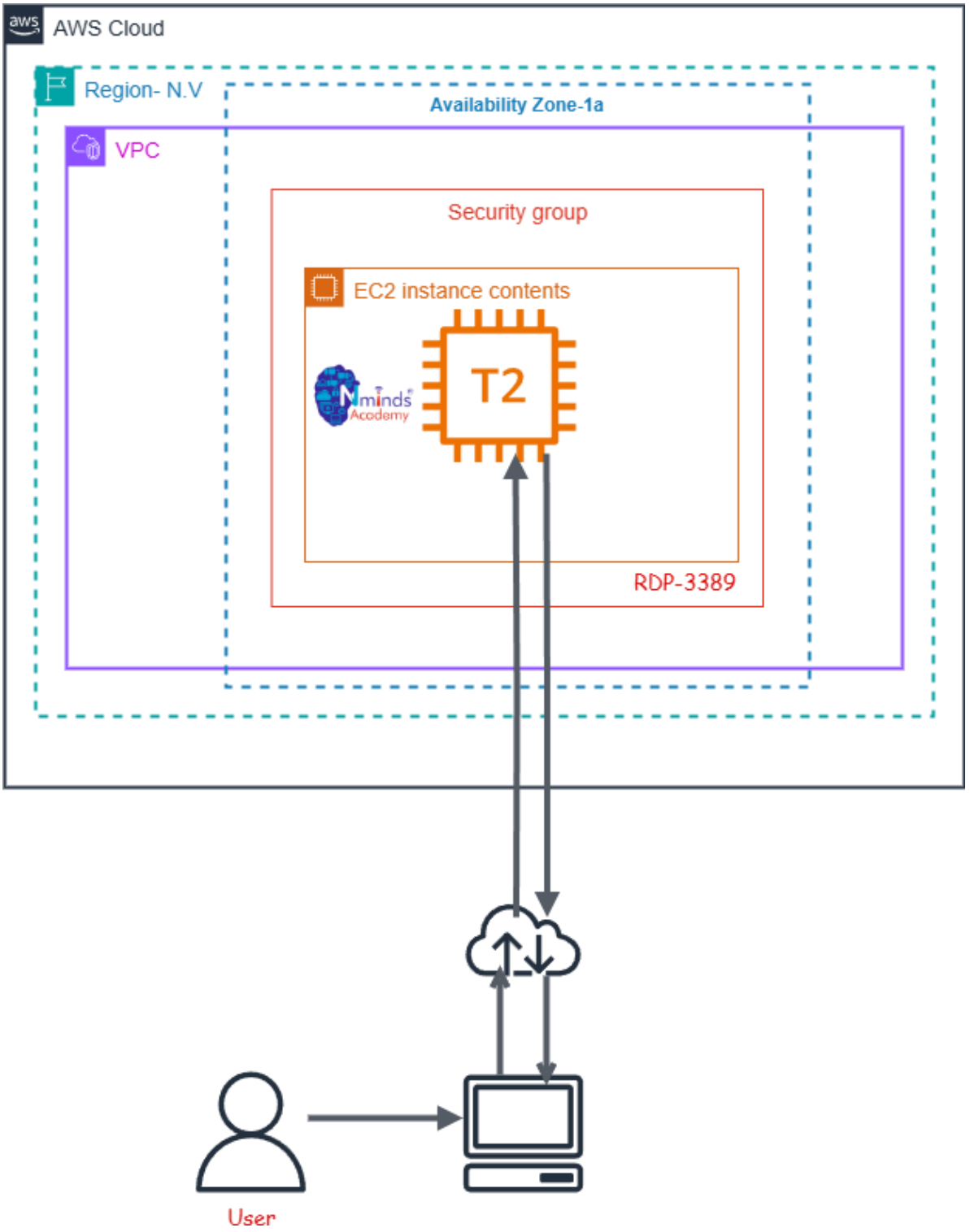
- Web and Application Hosting: Run IIS web servers or other Windows-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Windows instances, using RDS or EC2 for more control.
- Remote Desktop Applications: Set up RDS (Remote Desktop Services) for remote access to Windows desktops in the cloud.
- Development and Testing: Create development environments with Windows OS to test .NET or other Windows-based applications.

Used Services in AWS:

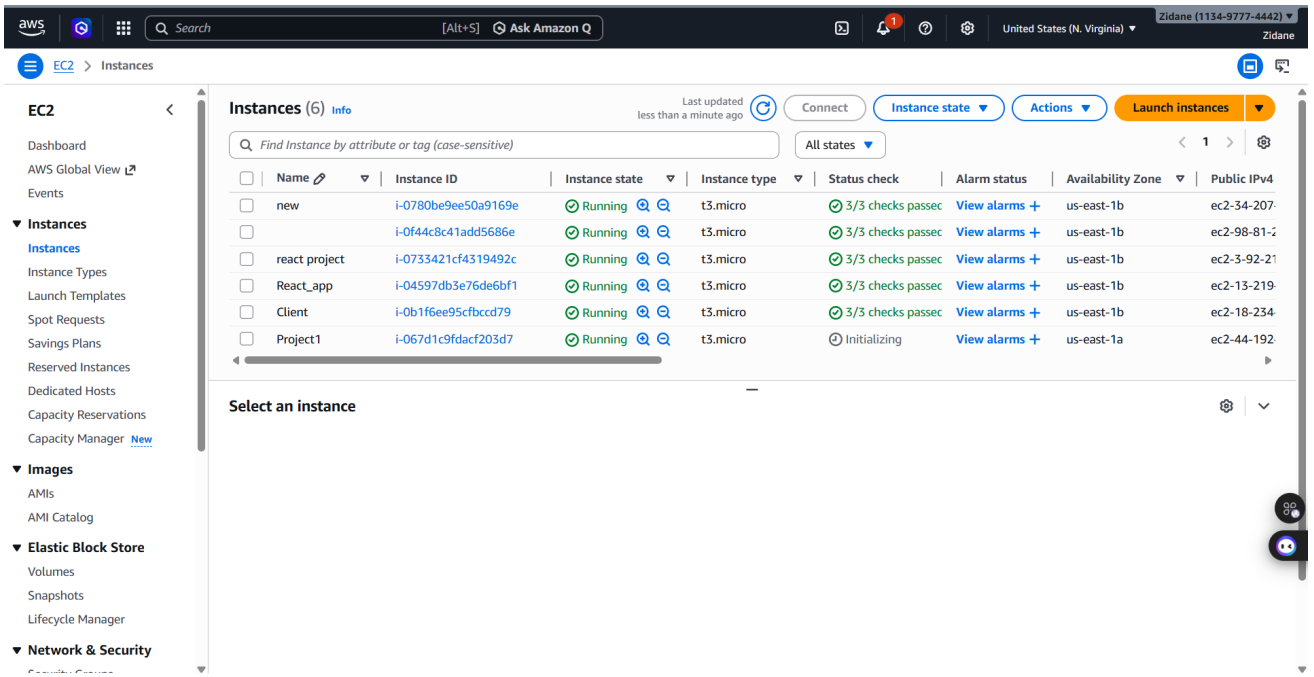


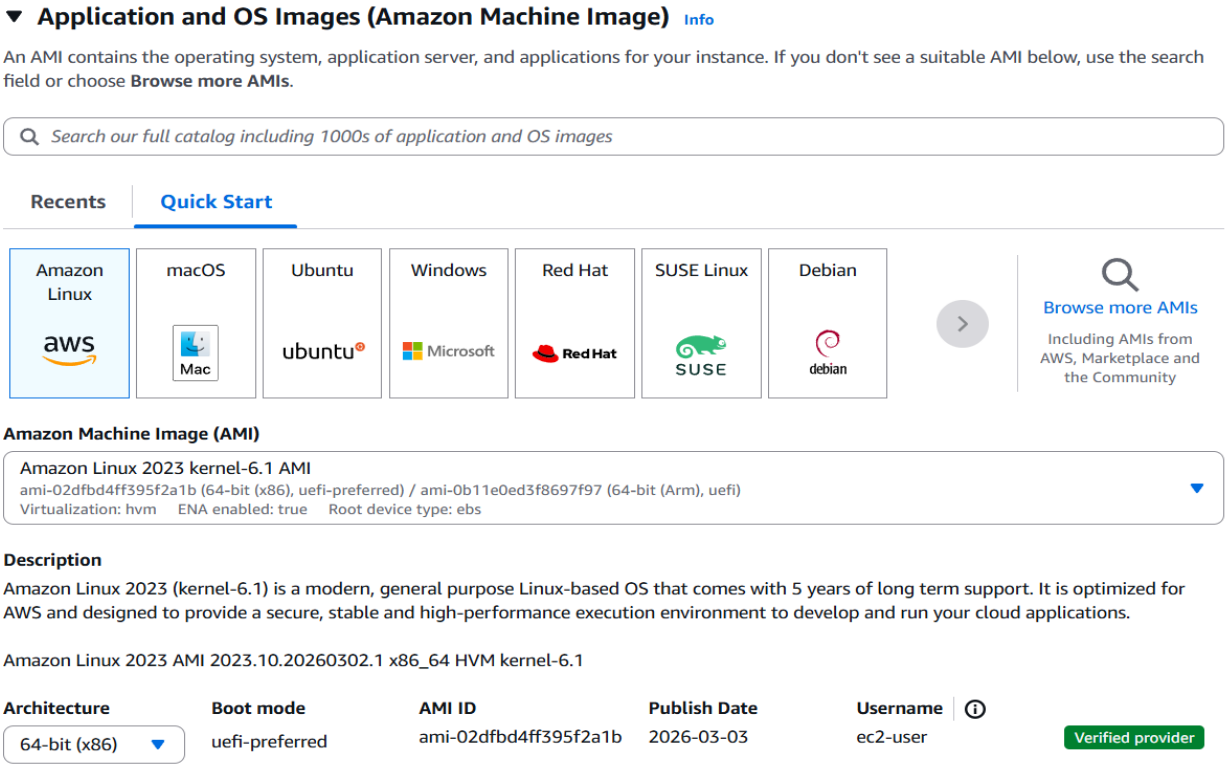
Amazon EC2

Topology


AWS Cloud
Region- N.V
Availability Zone-1a
VPC
Security group
EC2 instance contents
T2
RDP-3389
User

Execution Tasks:





Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

project1 | Create new key pair

Network settings Info

Edit

Network | Info

vpc-01983130ed784d7f5

Subnet | Info

No preference (Default subnet in any availability zone)

Auto-assign public IP | Info

Enable

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- (•) Create security group
- ( ) Select existing security group

We'll create a new security group called '**launch-wizard-7**' with the following rules:

- [x] Allow SSH traffic from — Anywhere 0.0.0.0/0
  Helps you connect to your instance
- [ ] Allow HTTPS traffic from the internet
  To set up an endpoint, for example when creating a web server
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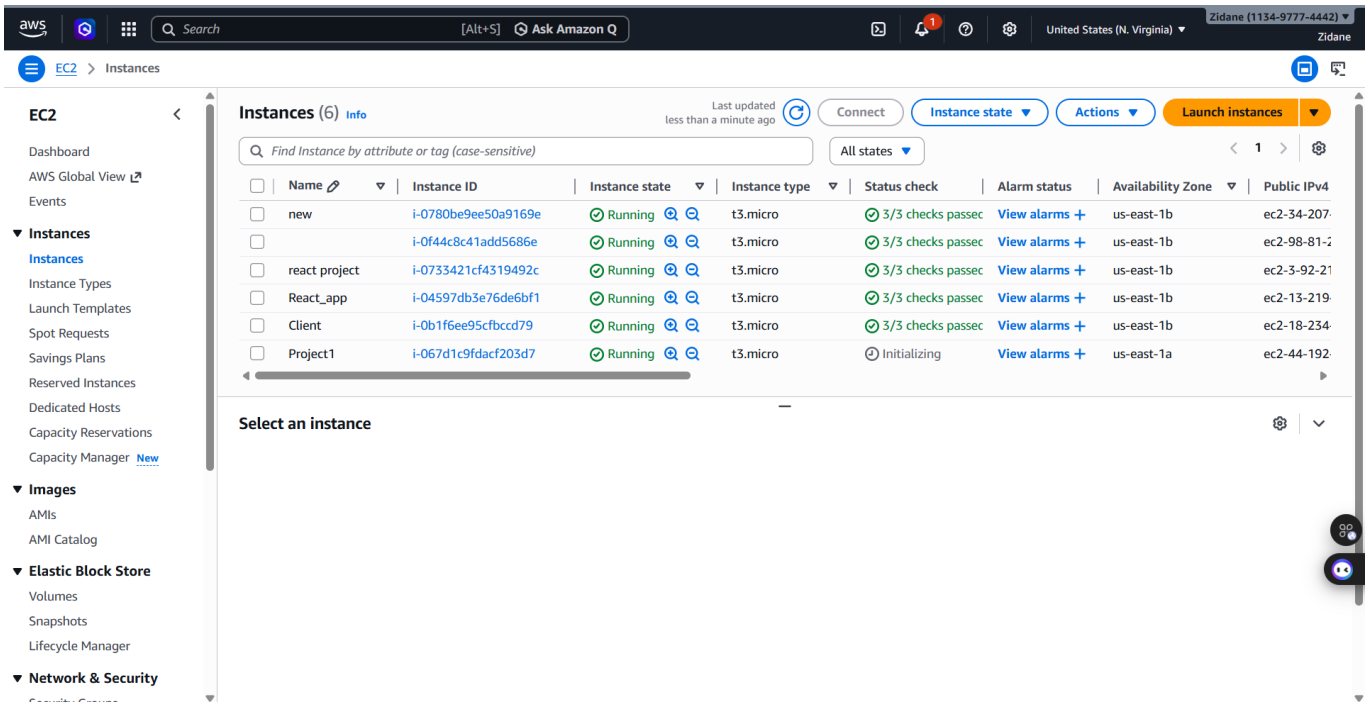
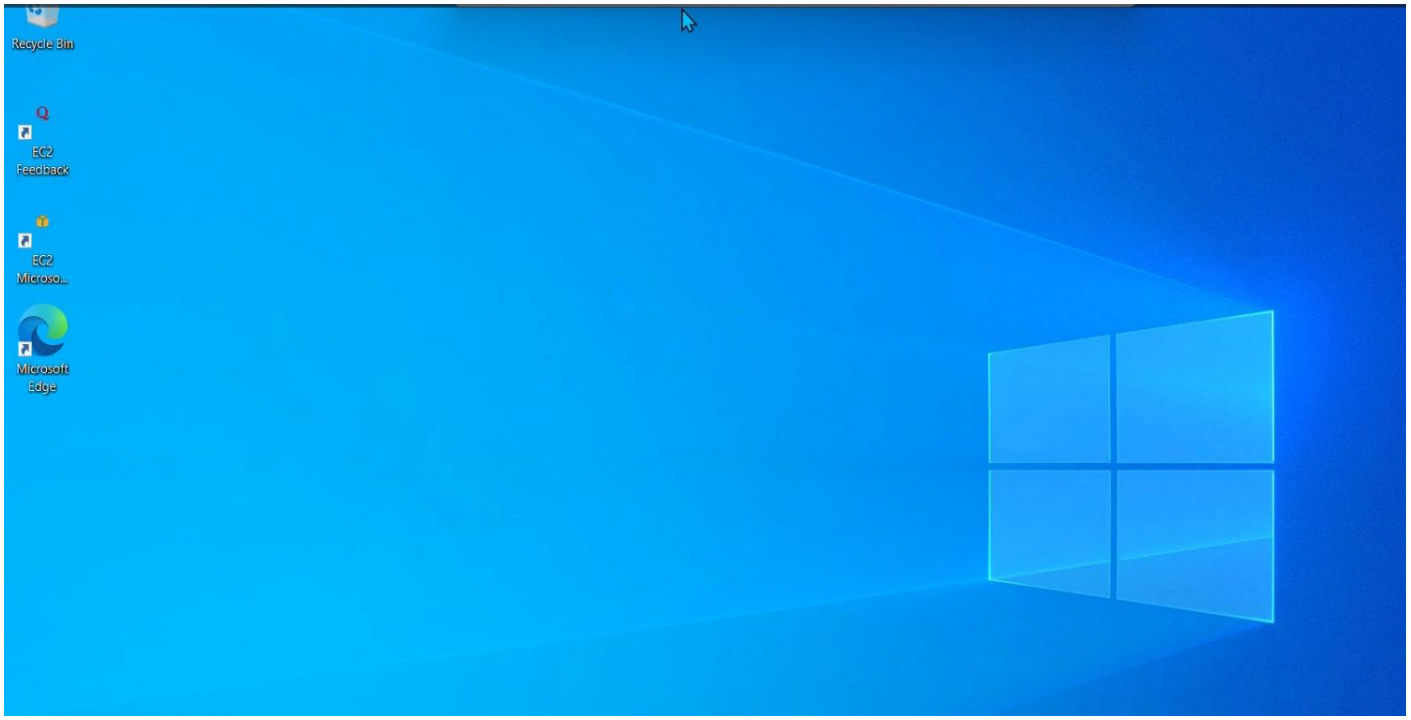
- [ ] Allow HTTP traffic from the internet
  To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ✕


aws
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Zidane (1134-9777-4442)
Zidane
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Instances (6) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
new i-0780be9ee50a9169e Running t3.micro 3/3 checks passed View alarms us-east-1b ec2-34-207
i-0f44c8c41add5686e Running t3.micro 3/3 checks passed View alarms us-east-1b ec2-98-81-
react project i-0733421cf4319492c Running t3.micro 3/3 checks passed View alarms us-east-1b ec2-3-92-21
React_app i-04597db3e76de6bf1 Running t3.micro 3/3 checks passed View alarms us-east-1b ec2-13-219
Client i-0b1f6ee95cfbccd79 Running t3.micro 3/3 checks passed View alarms us-east-1b ec2-18-234
Project1 i-067d1c9fdacf203d7 Running t3.micro Initializing View alarms us-east-1a ec2-44-192
Select an instance

Administrator
Please wait for the User Profile Service


Recycle Bin
EC2 Feedback
EC2 Microso...
Microsoft Edge

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2. Configure IIS Webserver on Windows Server in AWS

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Objective

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

IIS (Internet Information Services) is a web server software created by Microsoft for hosting websites and web applications on Windows Server operating systems. It provides a platform for building, deploying, and managing web-based applications. IIS supports multiple protocols, including HTTP, HTTPS, FTP, and more, and is tightly integrated with the Windows OS.

Common Use Cases of IIS:

1. Hosting Websites: IIS is often used by businesses and developers to host websites and web applications, particularly those built with ASP.NET.
2. Web Applications: For applications that require robust back-end support, IIS provides features like application pooling and security for web services (e.g., SOAP or RESTful APIs).
3. Secure File Sharing: IIS can be configured as an FTP server to securely share files between different locations.
4. Application Hosting in Enterprises: Many enterprise applications built on Microsoft technologies rely on IIS for deployment, such as SharePoint, Exchange Server, or Dynamics CRM.
5. API Hosting: IIS is commonly used to host APIs (e.g., RESTful services) built with .NET or other technologies.

Used Services in AWS

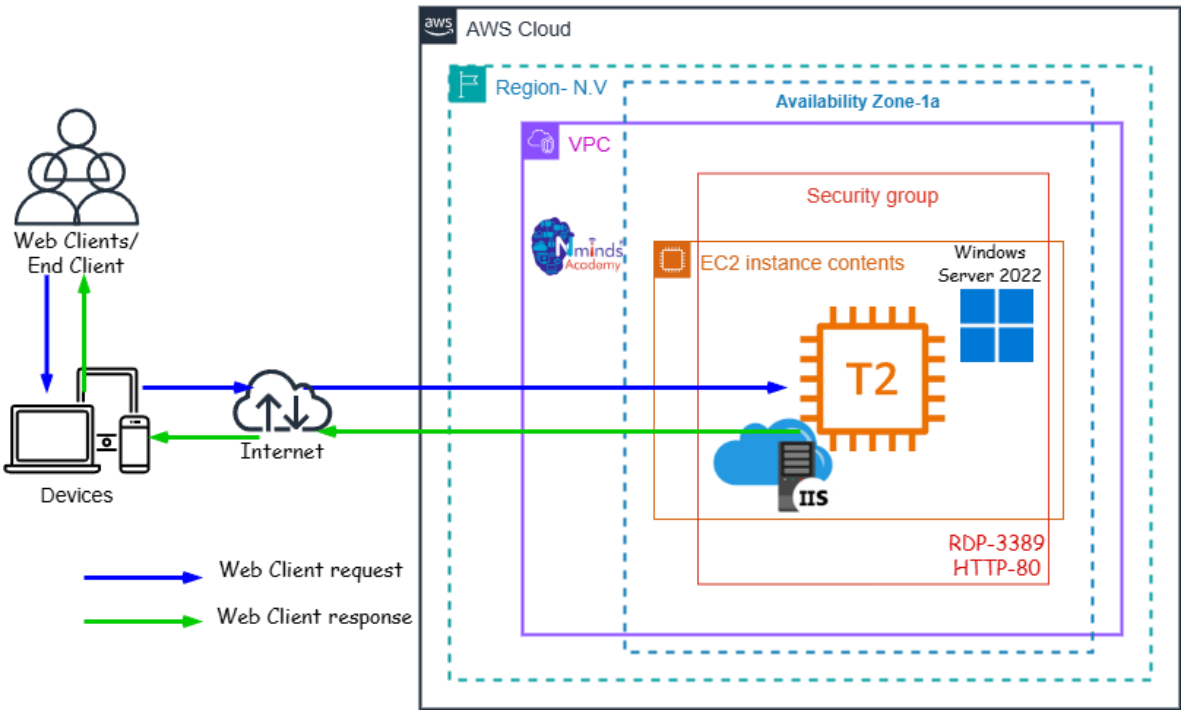
Used Services in Windows Server

Amazon EC2

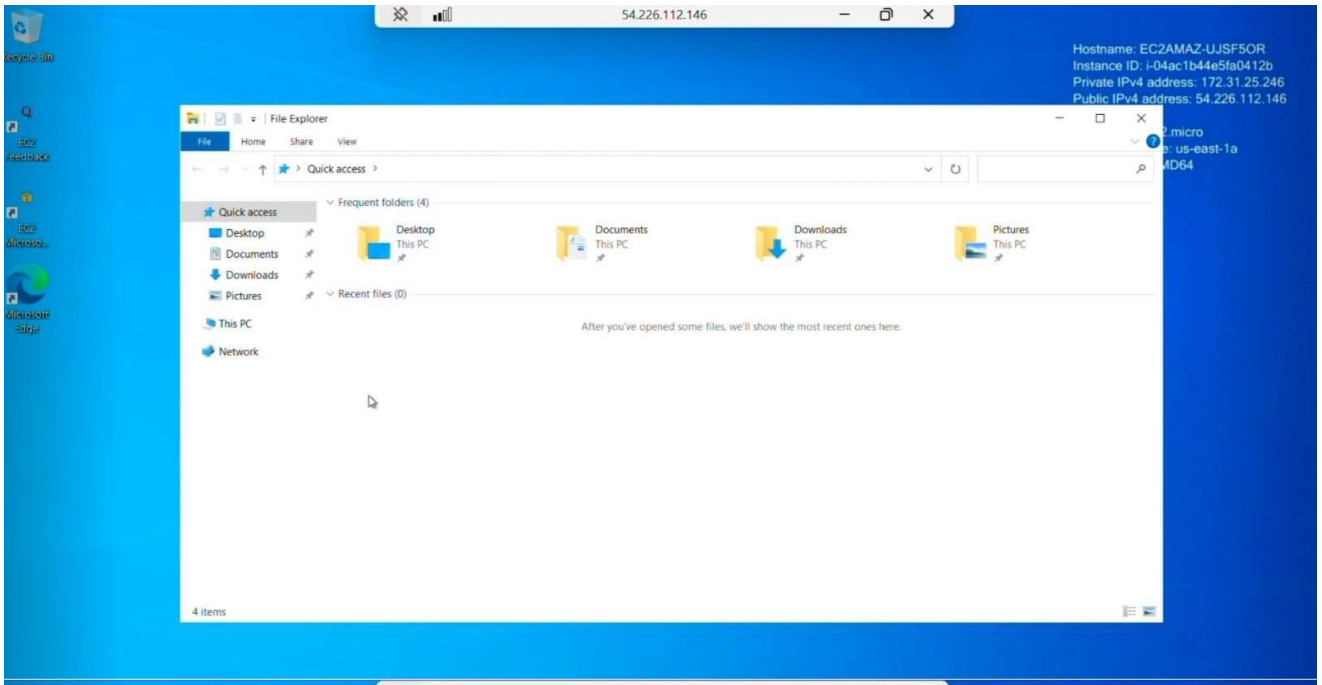
IIS- Internet Information Services

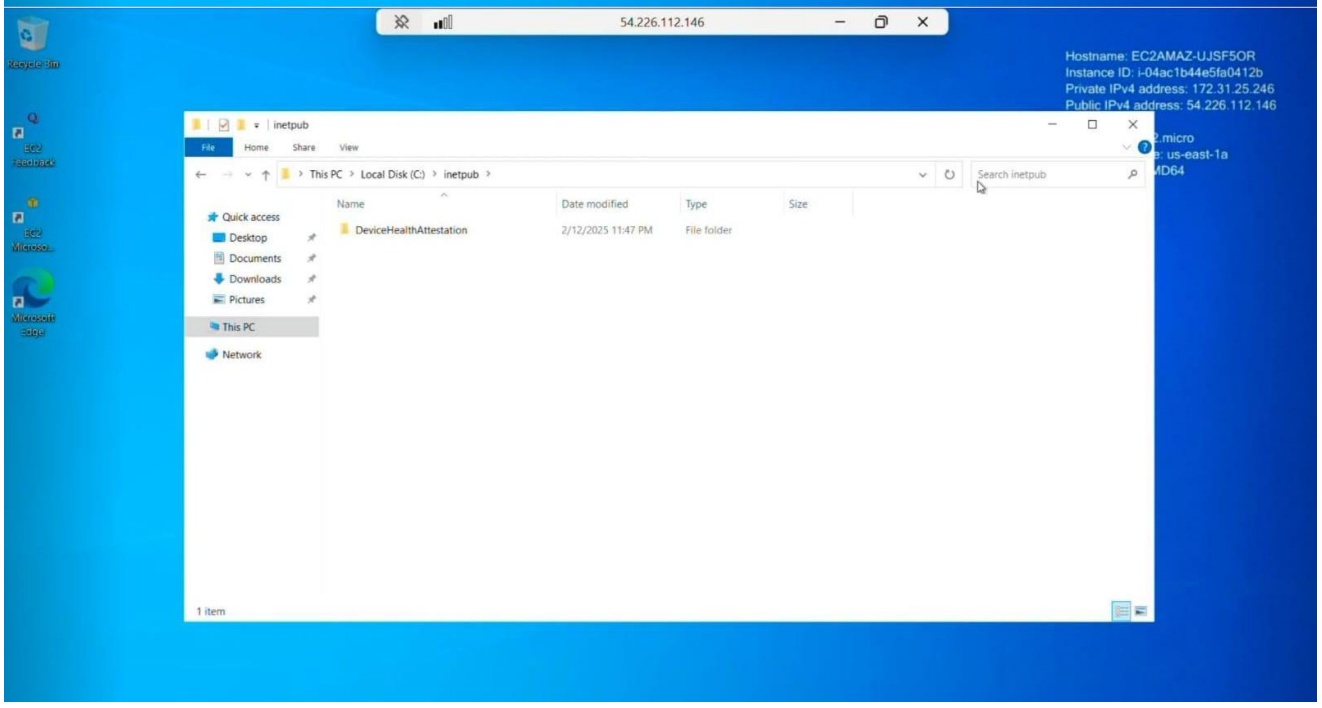
Topology

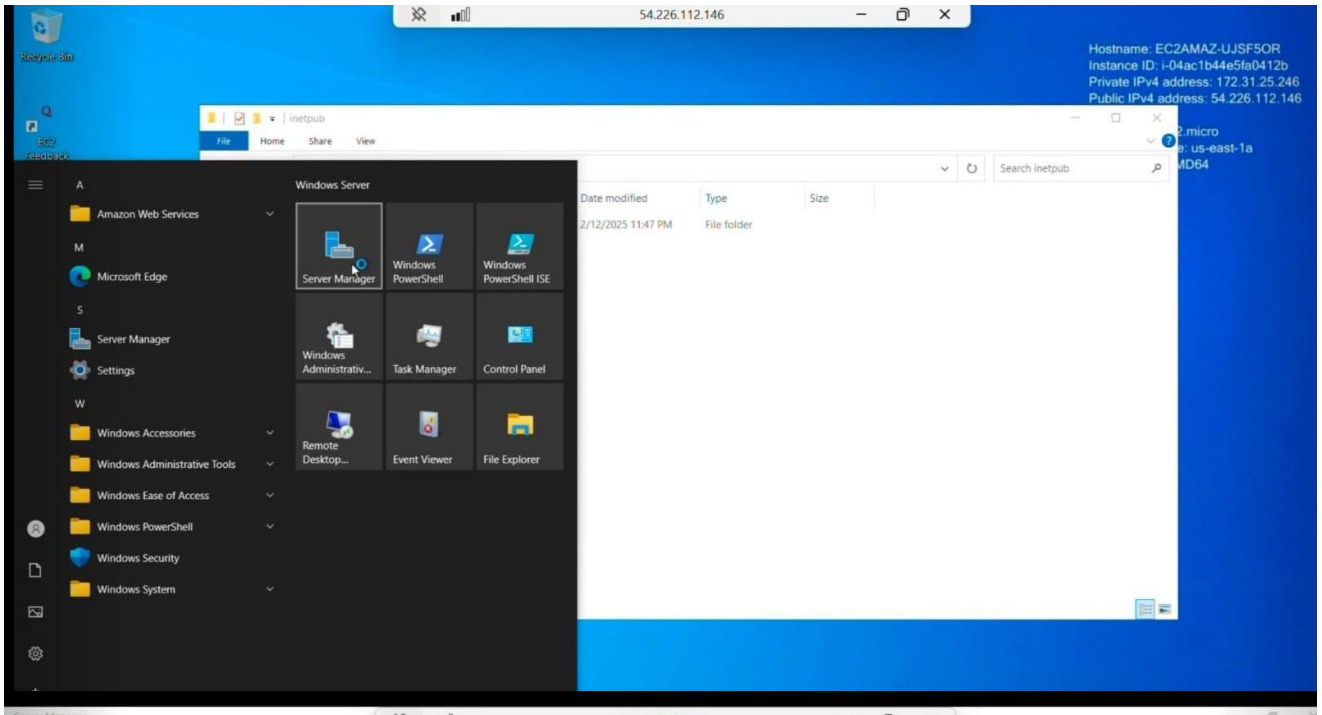
How to Configure the IIS web server in Windows Server with AWS EC2

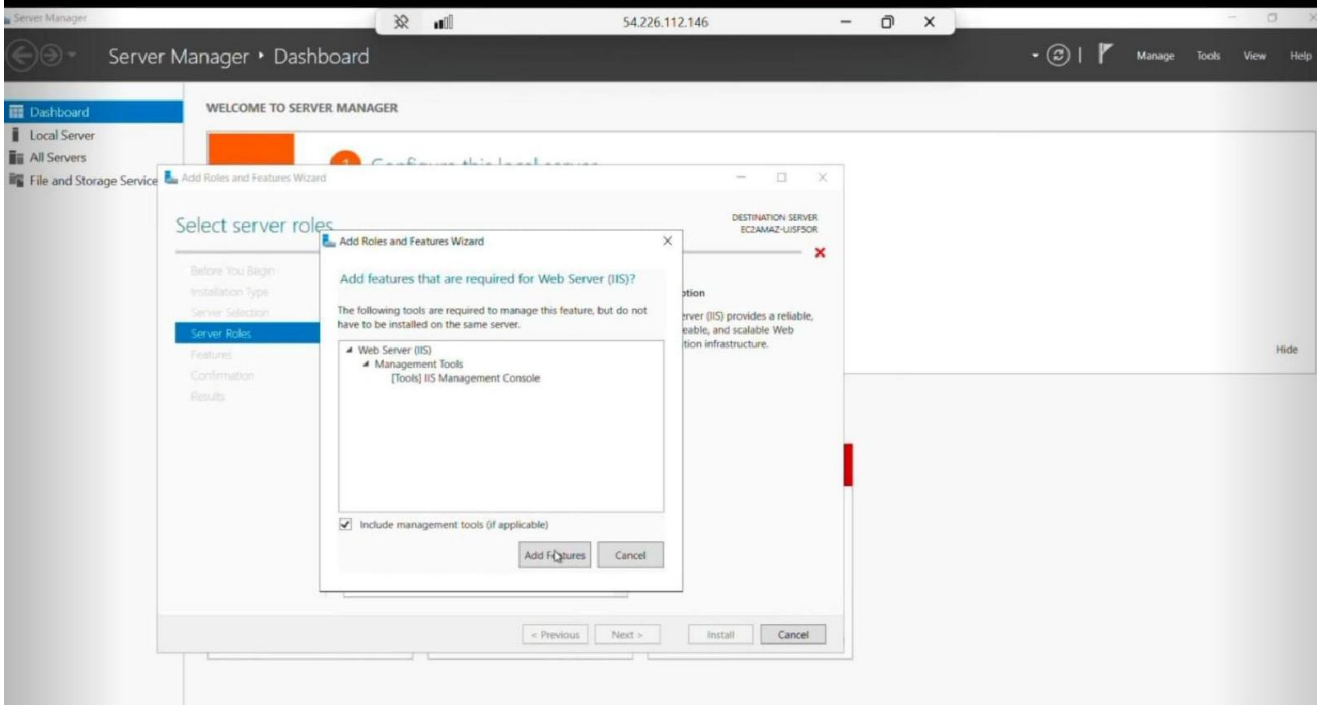


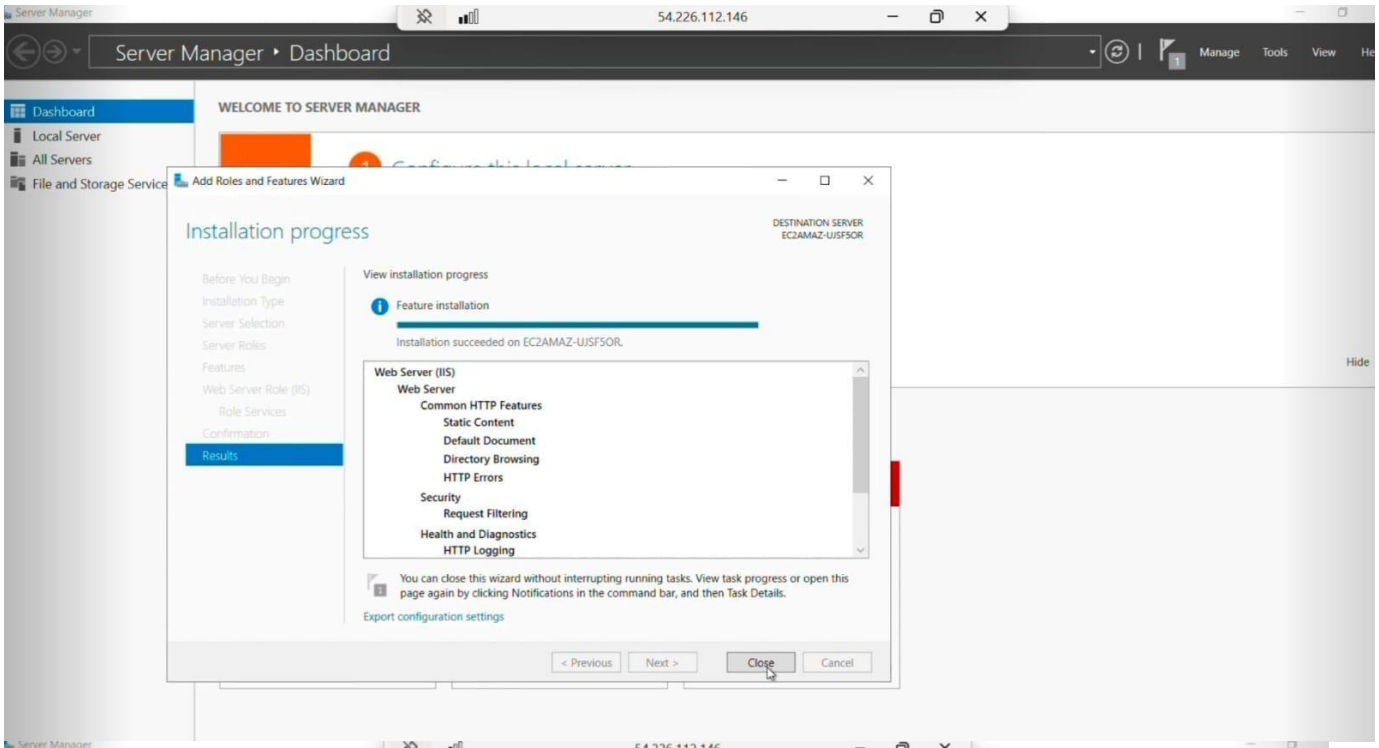
Execution Tasks:

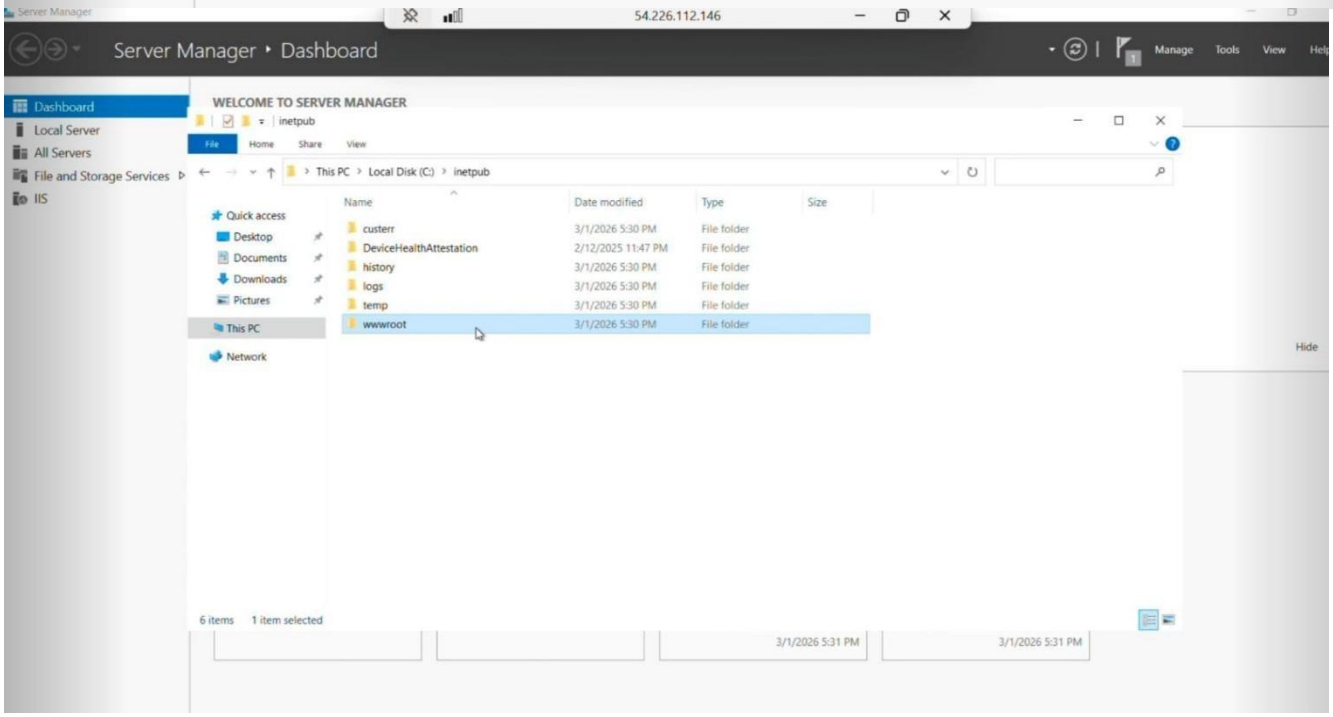


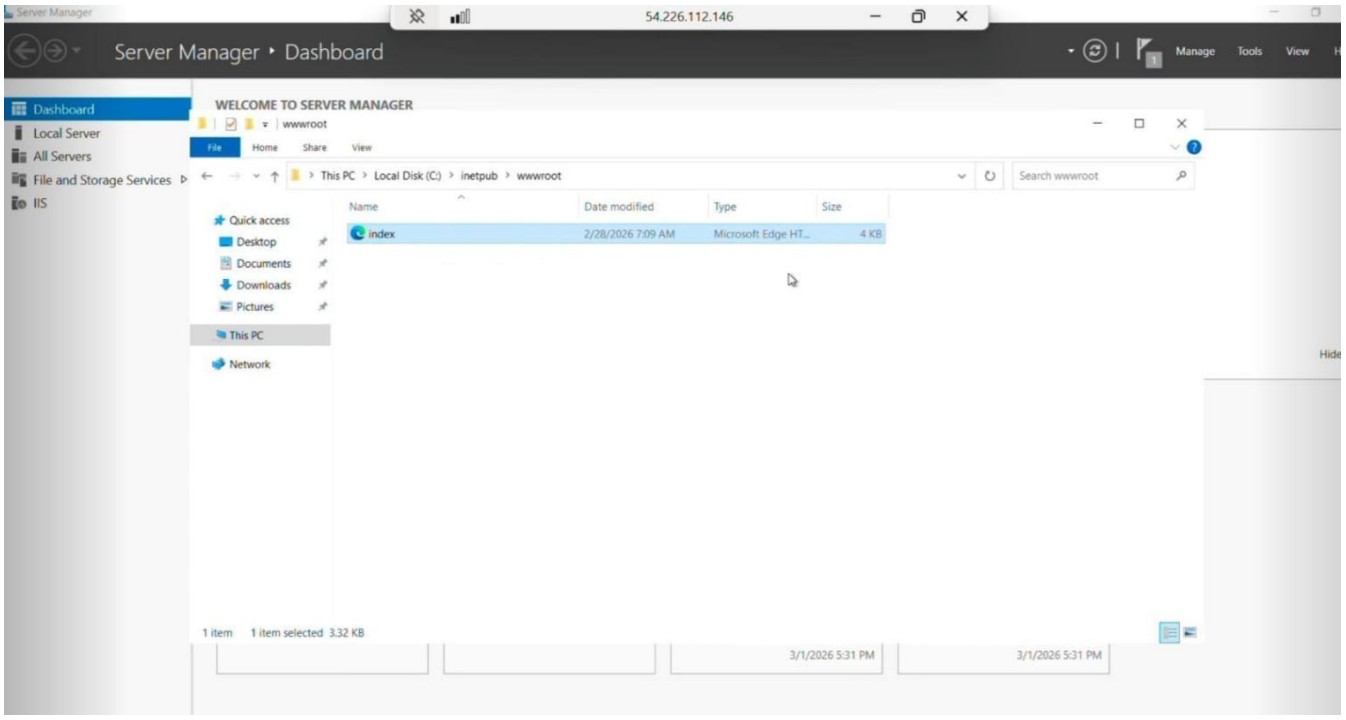


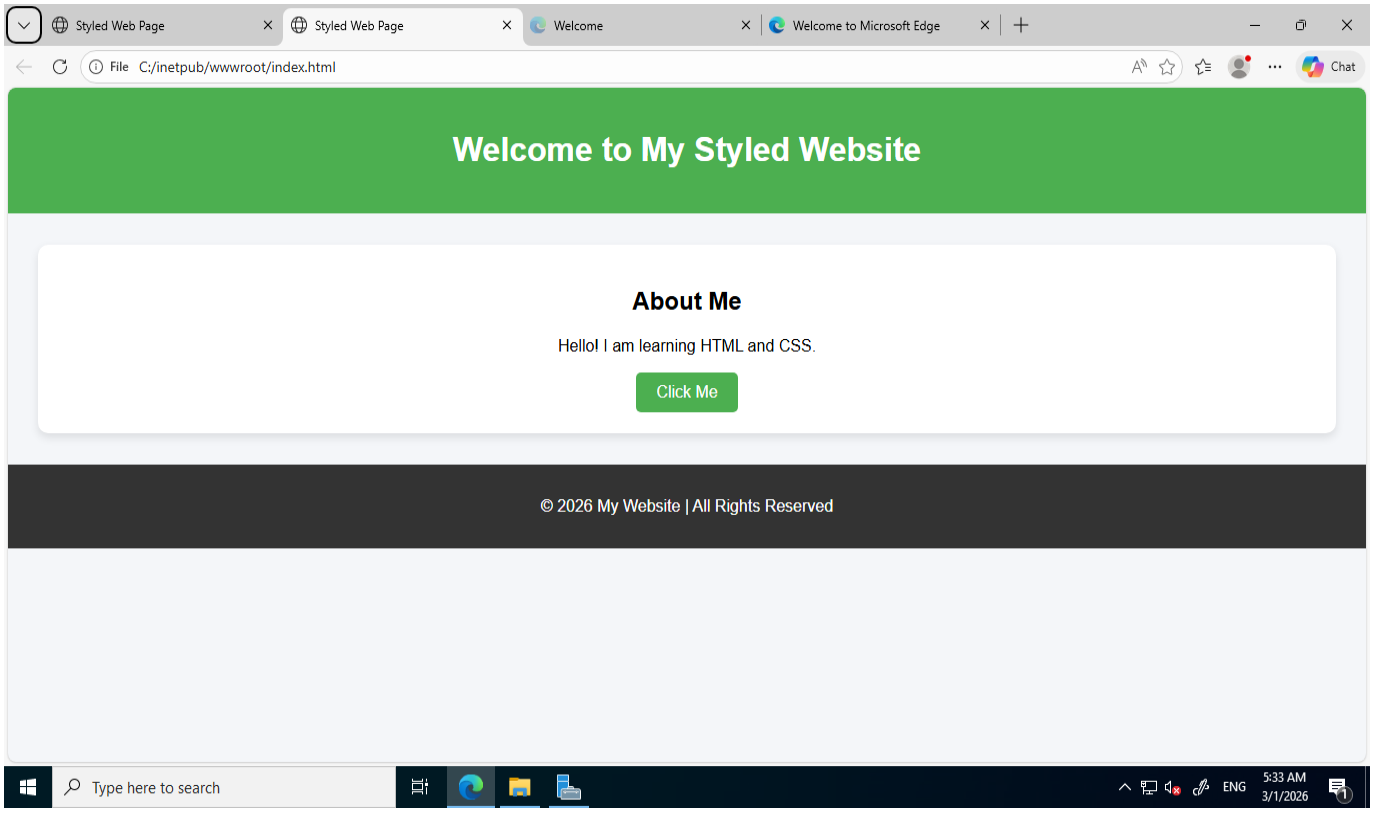

54.226.112.146
Hostname: EC2AMAZ-UJSF5OR
Instance ID: i-04ac1b44e5fa0412b
Private IPv4 address: 172.31.25.246
Public IPv4 address: 54.226.112.146
Recycle Bin
inetpub
File
Home
Share
View
Search inetpub
Date modified
Type
Size
2/12/2025 11:47 PM
File folder
Windows Server
Amazon Web Services
Microsoft Edge
Server Manager
Settings
Windows Accessories
Windows Administrative Tools
Windows Ease of Access
Windows PowerShell
Windows Security
Windows System
Server Manager
Windows PowerShell
Windows PowerShell ISE
Windows Administrativ...
Task Manager
Control Panel
Remote Desktop...
Event Viewer
File Explorer


54.226.112.146
Server Manager ‣ Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
File and Storage Service
WELCOME TO SERVER MANAGER
Add Roles and Features Wizard
Select server roles
DESTINATION SERVER
EC2AMAZ-UJSF5OR
Add features that are required for Web Server (IIS)?
The following tools are required to manage this feature, but do not have to be installed on the same server.
Web Server (IIS)
Management Tools
[Tools] IIS Management Console
Include management tools (if applicable)
Add Features
Cancel
Server Roles
< Previous
Next >
Install
Cancel
Hide


Server Manager
54.226.112.146
Server Manager ‣ Dashboard
Manage
Tools
View
Dashboard
Local Server
All Servers
File and Storage Service
WELCOME TO SERVER MANAGER
Add Roles and Features Wizard
Installation progress
DESTINATION SERVER
EC2AMAZ-UJSF5OR
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Web Server Role (IIS)
Role Services
Confirmation
Results
View installation progress
Feature installation
Installation succeeded on EC2AMAZ-UJSF5OR.
Web Server (IIS)
Web Server
Common HTTP Features
Static Content
Default Document
Directory Browsing
HTTP Errors
Security
Request Filtering
Health and Diagnostics
HTTP Logging
You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.
Export configuration settings
< Previous
Next >
Close
Cancel
Hide


Server Manager
54.226.112.146
Server Manager ‣ Dashboard
Manage
Tools
View
Dashboard
Local Server
All Servers
File and Storage Services
IIS
WELCOME TO SERVER MANAGER
inetpub
File
Home
Share
View
This PC ‣ Local Disk (C:) ‣ inetpub
Quick access
Desktop
Documents
Downloads
Pictures
This PC
Network
Name
Date modified
Type
Size
custerr
3/1/2026 5:30 PM
File folder
DeviceHealthAttestation
2/12/2025 11:47 PM
File folder
history
3/1/2026 5:30 PM
File folder
logs
3/1/2026 5:30 PM
File folder
temp
3/1/2026 5:30 PM
File folder
wwwroot
3/1/2026 5:30 PM
File folder
6 items
1 item selected
Hide


54.226.112.146
Server Manager • Dashboard
Manage
Tools
View
Dashboard
Local Server
All Servers
File and Storage Services
IIS
WELCOME TO SERVER MANAGER
wwwroot
File
Home
Share
View
This PC > Local Disk (C:) > inetpub > wwwroot
Search wwwroot
Quick access
Desktop
Documents
Downloads
Pictures
This PC
Network
Name
Date modified
Type
Size
index
2/28/2026 7:09 AM
Microsoft Edge HT...
4 KB
1 item
1 item selected 3.32 KB
3/1/2026 5:31 PM
3/1/2026 5:31 PM


Styled Web Page
Styled Web Page
Welcome
Welcome to Microsoft Edge
File C:/inetpub/wwwroot/index.html
Chat
Welcome to My Styled Website
About Me
Hello! I am learning HTML and CSS.
Click Me
© 2026 My Website | All Rights Reserved
Type here to search
ENG
5:33 AM
3/1/2026

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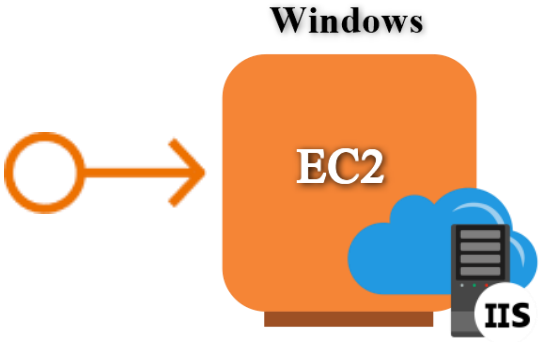
3. Configure Elastic IP Address to Windows Web Server in AWS



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Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

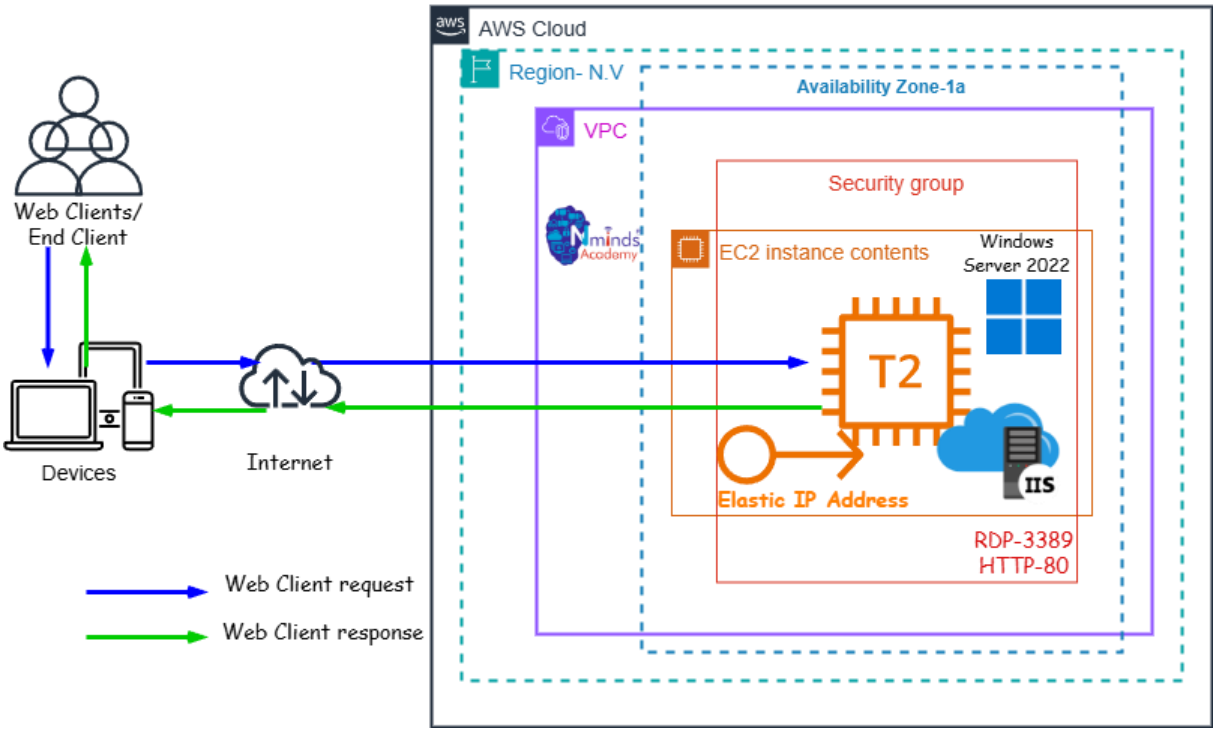
1. Highly Available Applications:
   - If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.
2. Web Servers:
   - If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.
3. Disaster Recovery:
   - Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

Best Practices:

- Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.

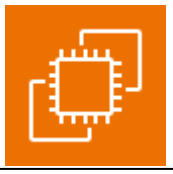
Topology

How to Configure the Elastic IP Address to Windows Web Server with AWS EC2



Used Services in AWS

Used Services in Windows Server

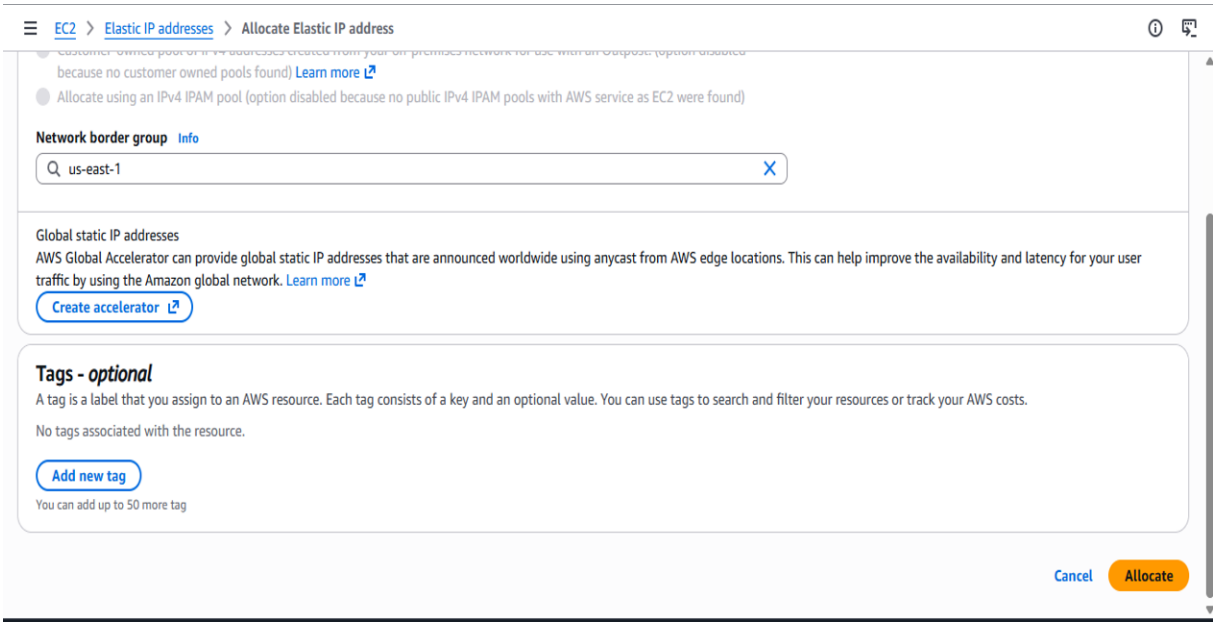


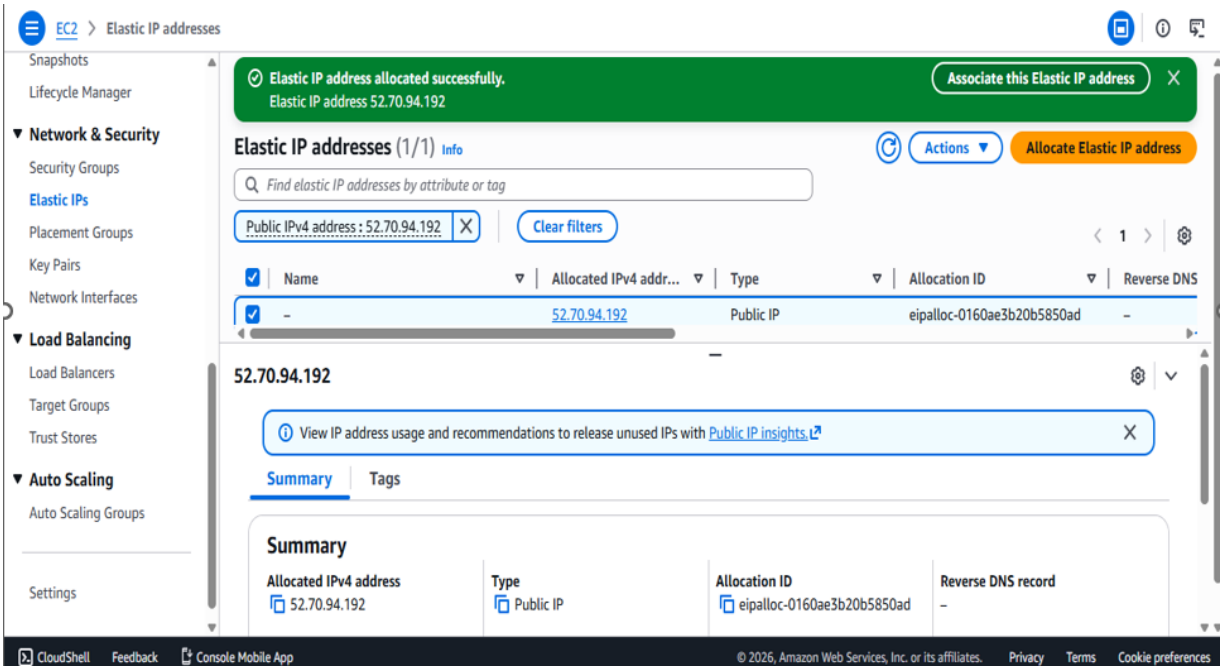
Amazon EC2

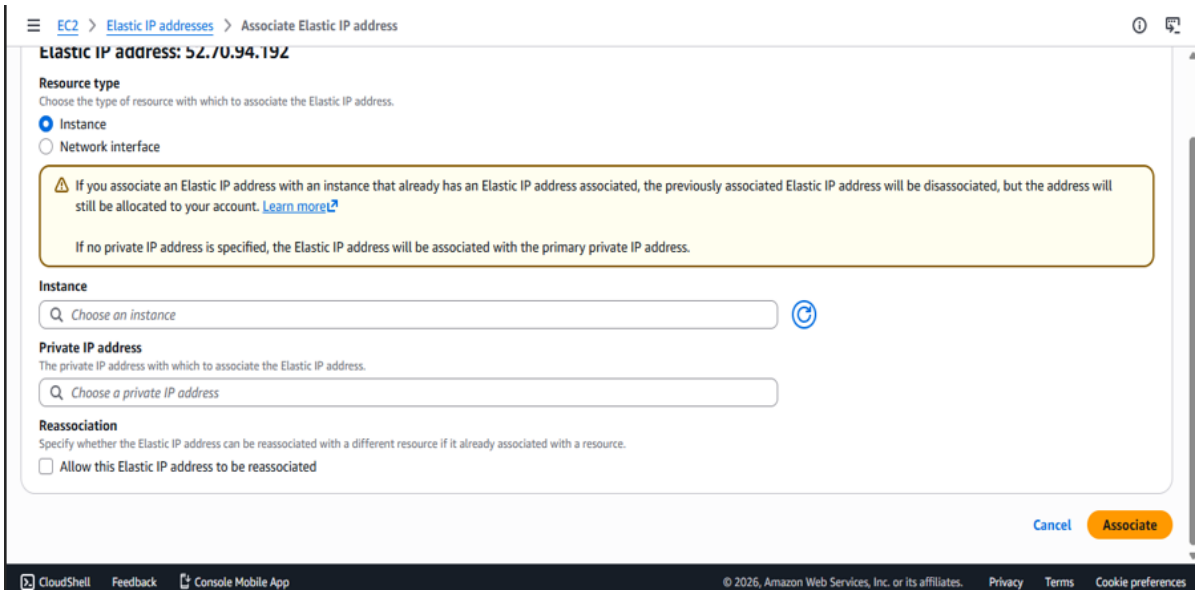
IIS- Internet Information Services

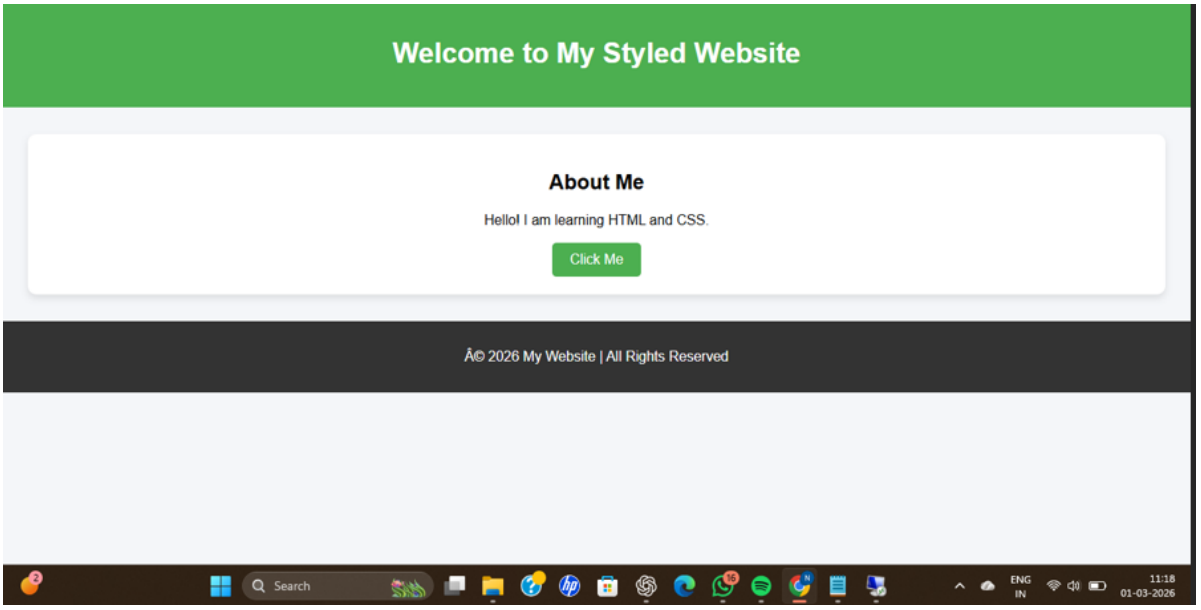
Amazon Elastic IP Address

Execution Tasks:






EC2 > Elastic IP addresses > Associate Elastic IP address
Elastic IP address: 52.70.94.192
Resource type
Choose the type of resource with which to associate the Elastic IP address.
Instance
Network interface
If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. Learn more
If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.
Instance
Choose an instance
Private IP address
The private IP address with which to associate the Elastic IP address.
Choose a private IP address
Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.
Allow this Elastic IP address to be reassociated
Cancel
Associate
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Welcome to My Styled Website
About Me
Hello! I am learning HTML and CSS.
Click Me
Â© 2026 My Website | All Rights Reserved
Search
ENG
IN
11:18
01-03-2026